After False Reassurance Spreads Network Recovery in an Evacuation Game

Hồ Lê Minh Thạch Bùi Minh Nhật Ngô Trần Bảo Thạch

Independent researcher (all authors)

With Apart Research

Abstract

Once false information spreads through a group, is removing its source enough? We test this in a simple evacuation game. Ten AI agents exchange local messages and decide whether to stay or leave. After a peer spreads a false claim, we copy the same game state and try different responses: do nothing, stop the source, or stop it and send a correction. We compare local corrections with a correction sent to everyone. We measure whether agents make the right decision and whether the false claim keeps spreading. This is an initial proposal with a completed small calibration batch and planned extensions. All recovery conditions start from the same saved conversation and actions, so earlier exposure does not vary between them. We include false reassurance during disaster and false warnings when there is no disaster. The main outcomes exclude the controlled source and distinguish actions taken before and after correction. An earlier single-model calibration batch provides descriptive checks, not confirmation of the broader proposed comparisons. Model, language, and identity extensions still require implementation and evaluation. The intended contribution is a reproducible test of recovery policies with matched starting conditions, while keeping message delivery separate from action and acknowledging that early evacuation cannot be reversed.

Scope note. This remains a separate recovery proposal. The revised paper A now tests language, claimed identity, and network structure; its three-experiment plan does not replace the recovery conditions below.

1. Introduction

1.1 Experiment C at a glance

Question: What helps a network recover after false information spreads?

Example: A peer says, "It is safe; stay here," when there is a disaster. Other agents can pass that claim along. We stop the original sender. Does the group still rely on its old message? Does a correction help?

Main comparison: Start from the same two rounds of conversation, then try five responses. Keep past messages in every version.

Success: Fewer missed evacuations during disasters and fewer unnecessary evacuations in safe games. A better explanation alone is not enough; actions must improve.

2. Related Work

Removing a sender does not remove what others have already heard. Agents may keep forwarding an old claim. Prior work already studies false testimony spreading through agents and persisting after its source exits [2]. We do not claim that this persistence is new. Our question is whether different recovery actions help when decisions have a cost and cannot be undone.

3. Methods

Fixed compute constraint. Use low reasoning effort for models and endpoints that expose and enforce that setting. Keep it unchanged across language, identity, and network conditions. Models without adjustable reasoning (including the GPT-4.1 mini calibration model) are recorded as not applicable, not as low. Exclude an endpoint from the comparable batch until its control is verified. Low is a provider-specific setting, not an equal reasoning-token budget across model families. Record output-token limits, token usage, model version, and provider separately.

3.1 The game in plain English

Ten AI agents must decide whether to stay or evacuate before a possible disaster. Each agent can talk only to four neighbours in a ring. One agent receives the true state from the simulator. The others must decide what to do using the messages they receive.

Evacuating costs one point but avoids the disaster. Staying earns two points if there is no disaster, and zero if disaster strikes. Surviving agents also earn 0.1 point for each other agent who takes the correct action. These points define the task; we do not assume that models care about rewards as humans do.

Each round follows the same steps: **read local messages, choose a message, choose stay or evacuate, then deliver messages for the next round.** Each agent has its own conversation history. Agents therefore respond repeatedly to one another; this is not a single prompt with a list of scripted opinions. Evacuation is final: an agent that leaves stops sending messages, but its old messages remain visible. In the calibration runs, the game ended after four to six rounds, without telling agents the exact final round.

Where the game comes from. Shirado, Crawford and Christakis used a human evacuation game with local Safe/Danger signals, costly evacuation, and an informed participant [1]. We borrow that setup. Turn-based play, AI agents, controlled claims, and our interventions are changes to the original experiment, not a direct replication.

3.2 What stays fixed and what changes?

Table 1. Fixed settings and proposed comparison factors.

Setting	Exact role in this proposal
Game rules: fixed	Ten agents; four neighbours each; the same points, evidence access, and irreversible evacuation rules.
Run settings: matched	Use the same scenario, source position, round schedule, instructions, and message limits when comparing treatments. Freeze generation settings before evaluation.
Disaster: balanced	Include both disaster and no-disaster games. Evacuating is not always the correct answer.
Claim accuracy: control	Test truthful and false claims. In C, the main recovery test uses false claims; honest-source games are an additional planned reference.
Communication: planned comparison	Compare the existing Safe/Danger signals with short natural-language messages that give reasons and can answer neighbours. Keep stay/evacuate decisions machine-readable. The text mode is not yet tested.
Model: planned comparison	Six named models are listed below: GPT-4.1 mini, Claude Haiku 4.5, Qwen3.8 27B, DeepSeek V4 Flash, Mistral Small 3.2, and Apertus 70B. Use one model for all agents in a game. Apertus access remains pending.
Language: planned comparison	Use the model-specific local language listed below: English, Chinese, French, or Standard German. Also compare every model in English and Vietnamese. Only English has been tested.
Source identity: planned comparison	Ordinary peer; emergency expert; political title or named figure such as Xi Jinping; religious role such as monk/pastor; named religious figure such as the Buddha; or a claim to speak for God. These are separate conditions, not one authority category. Only ordinary-peer labels have been tested.

The protocol compares six proposed models using shared English and Vietnamese conditions and additional model-specific languages. Appendix B gives the model roster, language assignments, and implementation details.

3.3 Exactly how experiment C works

1. Choose disaster or no disaster. Give one informant the truth. A separate controlled peer sends a false claim: Safe during a disaster, or Danger when there is none.

2. Let the agents exchange messages for two rounds. They may accept, question, ignore, or pass on the claim, and they may evacuate.

3. Save every history, delivered message, and action. Copy that exact game state into the five conditions below.

4. Continue each copy to the same scheduled end. Compare actions and messages after the intervention, as well as the final outcome of the whole game.

Table 2. Recovery interventions applied to a shared conversation state.

Condition	What changes after round two?
Do nothing	The source can keep sending.
Stop the source	Block future source messages. Keep its old messages.
Stop + local correction	Also send verified facts to the source's neighbours.
Stop + linked correction	Send the same facts to the same neighbours, adding IDs of the false claim and its recorded relays.
Stop + correction to everyone	Send the factual correction directly to every agent. This tests the benefit of wider delivery.

Use the same removal notice in all conditions that stop the source. Keep the correction's sender, verification, timing, and facts fixed. Keep the two local correction lengths similar. A correction sent to everyone reaches more agents by design; it is not an equal-message-budget comparison.

Why use a saved conversation? It ensures that each recovery condition starts after exactly the same exposure and actions. Apply the intervention at the fixed time even if the false claim has not spread. Selecting only successful misinformation attempts would change the question. For later identity tests, create separate exposure histories for each label, then compare recovery conditions within each history.

Planned natural-language version. Agents will exchange short reasons and replies as well as structured decisions. Keep IDs for messages and the messages they cite. An ID alone does not prove that a paraphrase preserves the same claim; review samples before using citations as a measure of false-information spread.

3.4 What we measure and compare

- **Missed danger:** What fraction of non-source agents stay during a disaster?
- **False alarms:** What fraction evacuate when there is no disaster?
- **Correction reach:** Who actually receives the correction, and what do they do next?
- **Continued spread:** Do agents keep passing on the false claim after the intervention? Also report points earned.

Compare local correction with source removal alone. Compare linked correction with the same local facts without claim links. Compare direct-to-all correction with local delivery. Pair outcomes from the same saved game; agents and copied continuations are not independent samples. Report uncertainty across original games, and report failed runs.

Count evacuations before correction separately. Leaving is irreversible, so a later correction cannot undo an early false alarm. Report both whole-game outcomes and decisions among agents still active when the correction arrives. Exclude the controlled false source from the main outcome denominator.

Additional checks still planned. Add an honest-source condition and removal of a comparable honest sender to separate misinformation effects from losing a communication link. Add games with no communication and games where everyone knows the truth to check basic task understanding. Removing a source changes network capacity as well as information, so its effect is not purely a test of correction.

4. Results

4.1 Calibration status; proposed evaluation pending

A small calibration batch used `openai/gpt-4.1-mini`, English, ordinary-peer labels, and structured Safe/Danger signals. It completed 129 of 130 planned outcomes across A and C; one A run failed message-reference validation. The repository contains the full calibration report. These runs checked the setup and produced descriptive results; they are not the planned model, language, identity, or natural-language comparison.

Natural-language arguments and the broader comparisons above remain planned. No results about Xi Jinping, the Buddha, God, languages other than English, or differences between model origins are claimed. The papers remain proposals. Before a larger evaluation, freeze prompts, model versions, message limits, sample sizes, stopping rules, and treatment order. Keep failed runs visible rather than silently dropping them.

5. Discussion and Limitations

The game is a simplified test of communication and action. A result on one ring network does not establish how all agent networks behave. Repeating the same model many times does not create many independent model families. Use whole games as the units of analysis, not individual agents, because agents influence each other. Use only simulated actions; external tools and hidden chain-of-thought are not needed.

The unpublished *Authority Across Languages* proposal informed the source-label and language questions. It is background, not evidence of experimental results. We make no claim that this is the first study of these topics.

6. Conclusion

This proposal defines comparisons to run, not findings from the revised evaluation. Its value must be assessed using the planned controlled experiments and reported failures as well as successful runs. No claim of proven safety improvement is made.

Code and Data

Project repository; scenario data in `data/evacuation/`; earlier calibration outcomes and analysis in `reports/2026-11-08-calibration/`. The existing runner and calibration artifacts do not implement or validate every proposed condition.

References

[1] H. Shirado, F. W. Crawford, and N. A. Christakis (2020). *Collective communication and behaviour in response to uncertain 'Danger' in network experiments*. Proceedings of the Royal Society A, 476:20190685. https://doi.org/10.1098/rspa.2019.0685.

[2] C. Yan et al. (2026). *When Truth Is Distributed: Misinformation Derails Collective Fact Recovery in LLM-Based Multi-Agent Systems*. arXiv:2608.03421v3. https://arxiv.org/html/2608.03421v3.

A. Limitations and Dual-Use / Ethical Considerations

The calibration describes decisions by simulated agents under a fixed payoff scheme. Copying a conversation allows comparison of corrections after the same exposure, but a correction cannot reverse an evacuation already taken. Final totals therefore mix preventable later errors with irreversible earlier decisions. These results cannot establish which response would protect people in an actual emergency.

Knowing which false claims survive source removal could help a malicious sender prolong their influence or exploit gaps in local corrections. Broadcasting also has a cost: a mistaken correction could reach agents who never saw the original claim. Evaluate missed danger and unnecessary evacuation separately, report who received each correction before acting, and keep sender removal and broadcasts confined to the simulator. These experiments provide no basis for suppressing real witnesses or issuing public evacuation advice.

B. Model and Language Protocol

B.1 Named models for the planned comparison

We propose the following six-model roster for both studies. This is a list of specific test models, not a list of all models available in each region. Only GPT-4.1 mini has completed the calibration batch. The other five need a task-comprehension and output-format check before evaluation.

Table 3. Proposed model roster and access status.

Developer / location	Model, exact identifier, and status
OpenAI / US	GPT-4.1 mini: `openai/gpt-4.1-mini`. OpenRouter; calibration completed.
Anthropic / US	Claude Haiku 4.5: `anthropic/claude-haiku-4.5`. OpenRouter; planned.
Alibaba Qwen / China	Qwen3.8 27B: `qwen/qwen3.8-27b`. OpenRouter; planned.
DeepSeek / China	DeepSeek V4 Flash (0423): `deepseek/deepseek-v4-flash`. OpenRouter; planned.
Mistral AI / France	Mistral Small 3.2 24B Instruct: `mistralai/mistral-small-3.2-24b-instruct`. OpenRouter; planned.
Swiss AI Initiative / Switzerland	Apertus 70B Instruct (2509): `swiss-ai/Apertus-70B-Instruct-2509`. Hugging Face checkpoint; inference endpoint still to be arranged.

The first five IDs were checked against the OpenRouter model catalogue on November 8, 2026. Apertus is identified by its official model card; it was not found under the Swiss AI namespace in that catalogue. A catalogue entry is not a successful test call. We will record the resolved model version and serving provider for each run, and freeze reasoning settings as well as sampling settings. These models differ in age, size, and training, so this is not a capability-matched test of regions.

Use the same model for all ten agents within a game, then rerun the matched scenarios with the next model. If Apertus cannot be served within the pilot budget, report a five-model comparison and mark Apertus as pending. Do not silently substitute a second Mistral model or claim two independent European developers were tested.

B.2 Which language does each model use?

The main comparison uses a language associated with each developer's location, as listed below. We call this the *local-language condition*. Models do not have a human native language, and several locations have more than one language. For Switzerland, Standard German is our explicit study choice, not a claim that it is Apertus's only or best language.

Table 4. Language assignments for each model.

Model	Local-language condition	Shared baseline	Extra language
GPT-4.1 mini	English	English (same condition)	Vietnamese
Claude Haiku 4.5	English	English (same condition)	Vietnamese
Qwen3.8 27B	Simplified Chinese	English	Vietnamese
DeepSeek V4 Flash	Simplified Chinese	English	Vietnamese
Mistral Small 3.2	French	English	Vietnamese
Apertus 70B	Standard German	English	Vietnamese

What changes with language? Translate the task instructions, evidence, controlled claims, source labels, correction messages, and agent-facing conversation. Ask all agents to write their natural-language messages in that game's language. Keep machine-readable action keys and message IDs unchanged. Check translation meaning before evaluation, and record when an agent uses the wrong language or mixes languages. Do not silently count such runs as valid language-following examples.

Why retain English for every model? Comparing a Chinese model in Chinese with a French model in French changes two things at once. That comparison describes the combined model-and-language setup. Running both in English gives a shared baseline. Comparing the same model in English and its local-language condition tests language sensitivity within that model. English is counted only once for the US models.

Where does Vietnamese fit? Vietnamese is a separate shared-language test across all six models, including those whose developers are outside Vietnam. Reuse the same scenarios and treatments. This tests performance in Vietnamese; it does not make any model Vietnamese or measure Vietnamese culture. With all six models available, there are 16 distinct model–language combinations: six English, six Vietnamese, and four additional local-language conditions. This is a list of planned conditions, not a commitment to run every identity and treatment combination at once.

How to test identity fairly. Change the claimed sender label on the controlled message while keeping its words, evidence, timing, and network position the same. The label gives no extra permissions or true information. These are fictional attributions in the game, not real statements by those people or religious figures. Keep the correction sender fixed in C. Record refusal or role-play responses separately. Freeze exact labels and translations before running the comparison.

How to test models and languages fairly. Run the same scenarios and treatments for every selected model. Record the developer, exact model ID, access date, API host, and generation settings. A hosting provider is not the model's developer. Geographic labels describe the sample; they do not explain why a model behaves differently. Model ability, training, size, and language skill can all differ. We cannot infer national culture, political loyalty, or religious belief from these results.

Keep the first comparison small. First check the natural-language mode in English with an ordinary-peer source. Then run the named models in the listed local-language conditions and shared English baseline. Add Vietnamese and source labels in separate, clearly listed batches. Do not change all three at once and then guess which caused a difference. Choose the number of games, repeated runs, message length, and failure rules before each evaluation batch. Mixed-model groups and changes to the network shape need separate designs.

LLM Usage Statement

An AI assistant helped investigate sources, design comparisons, write simulator code, and draft and edit this report. The recorded calibration outputs were checked by the repository's validators. Human authors must review the source interpretations, methods, and final text; independent human verification of every claim is not asserted.